



JsPL

*Write a library in three languages
simultaneously!*

Website: <https://jspl.merzin.dev>

Source: <https://github.com/rasmusmerzin/jspl>

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JsPL is a JavaScript-Python-Lua 3-way transpiler made using Tree-sitter, a powerful parser generator tool that enables accurate syntax tree construction for code analysis and transformation. This project aims to facilitate seamless code conversion between these three popular programming languages by leveraging Tree-sitter's robust grammars to parse, analyze, and generate equivalent syntax across language boundaries. By creating a unified intermediate representation and implementing precise code generation rules, this transpiler will help developers prototype across ecosystems or explore polyglot programming with greater ease.

Hello World Example

HelloWorld.js

```
function main() {
  print("Hello, JsPL!\n");
}
```



HelloWorld.py

```
def main():
  print("Hello, JsPL!\n")
```



HelloWorld.lua

```
function main()
  print("Hello, JsPL!\n")
end
```

Method Call Example

DotAccessor.js

```
function dotAccessor(a, b) {
  let text = Math.floor(a, b).toString();
  System.out.println(text);
}
```



DotAccessor.py

```
def dotAccessor(a, b):
  text = Math.floor(a, b).toString()
  System.out.println(text)
```



DotAccessor.lua

```
function dotAccessor(a, b)
  local text = Math:floor(a, b):toString()
  System.out:println(text)
end
```

While Loop Example

While.js

```
function whileLoop(count) {
  while (count > 0) {
    let i = 0;
    while (i < count) {
      print(" x");
      i = i + 1;
    }
    print("\n");
    count = count - 1;
  }
}
```



While.py

```
def whileLoop(count):
  while count > 0:
    i = 0
    while i < count:
      print(" x")
      i = i + 1
    print("\n")
    count = count - 1
```



While.lua

```
function whileLoop(count)
  while count > 0 do
    local i = 0
    while i < count do
      print(" x")
      i = i + 1
    end
    print("\n")
    count = count - 1
  end
end
```